Greenfoot Project “Shell Out!”:
Game Instruction Document

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Game Instruction

Shell Out! is an arcade-style obstacle-avoidance game inspired by Frogger. The player takes control of a parent turtle who must navigate across a busy road to rescue its three missing children and safely return home. The road is filled with moving vehicles traveling at different speeds and directions, making each crossing a challenge. Players must avoid traffic, collect all three baby turtles, and reach home before a sixty-second countdown expires. A flock of ravenous pelicans threatens the baby turtles, adding urgency to the rescue mission. The objective is to rescue all three baby turtles and successfully reach home before time runs out or the player loses all available lives.

Game Objective

The main objective of Shell Out! is to rescue all three baby turtles scattered throughout the road and safely guide the parent turtle to its home.

Players must:

- Navigate through moving traffic without being hit using the W, A, S, and D keys to move up, down, left, and right, respectively;
- Locate and collect all three baby turtles by colliding (or landing on top of) their sprites;
- Keep track of their remaining lives. They can only win if they keep their mistakes to a minimum;
- Collect all three baby turtles before the sixty-second timer reaches zero; and,
- Reach the home destination after collecting all three children.

The player must complete both parts of the objective (rescuing all three children and reaching home) in order to win the game. Partial completion will not result in a win state, and the timer will continue to decrement.

Player and Game Actors

The player operates a single turtle, but the game world also includes other actors, or characters, including the baby turtles, vehicles, the ravenous pelicans, and home.

Player Turtle

The Player Turtles is the main character controlled by the player. It is represented by a cartoon-style green turtle with a green shell. The player must guide this turtle through traffic, rescue its children, and safely reach home. The Player Turtle responds to the WASD keyboard controls.



Baby Turtles
The Baby Turtles are three smaller green turtles scattered randomly throughout the road. They represent the player's missing children. The player must make contact (or collide) with each baby turtle to rescue it. Once collected, a baby turtle disappears from the game world, and the rescue counter increases.



Vehicles
Vehicles are the primary obstacles in the game. They are represented by various cartoon-style cars and trucks that travel horizontally across the road in designated lanes. Vehicles move at different speeds and directions, requiring players to carefully time their movements and avoid collisions.





Ravenous Pelicans
The Ravenous Pelicans are the antagonists of the game. They are represented by cartoon-style pelicans with large beaks. If the sixty-second countdown reaches zero before the player completes the rescue, the pelicans enter the game world and steal the remaining baby turtles, resulting in a game over.



Home
Home is the player's final destination and the safe space where the turtle family can reunite. It is represented by a small grassy area on the opposite side of the road from the starting location. The player must reach this location after collecting all three baby turtles to win the game.



Controls and Gameplay
The Player Turtle is controlled using the keyboard's WASD keys. Each key moves the turtle in a specific direction. There are no other controls.

Key	**Action**
W	Move Up
A	Move Left
S	Move Down
D	Move Right

Gameplay Instructions

1. Begin the game with the Player Turtle at its starting location.
2. Use the WASD keys to navigate across the road.
3. Watch for vehicles traveling in different lanes and avoid collision with them.
4. Locate and collect all three Baby Turtles by making contact with them.
5. Keep an eye on the sixty-second countdown timer and children rescued.
6. Once all three children have been rescued, navigate to the Home location.
7. Reach Home before time expires to complete the game.

Players should move carefully and pay attention to the movement patterns of vehicles. Rushing across the road without considering traffic can result in lives lost.

Scoring and Lives
Players earn points by successfully rescuing Baby Turtles and reaching Home. Additional bonus points are awarded based on how quickly the player completes the rescue and how many lives remain after the rescue is complete.

The game tracks the following statistics:

- **Score:** The player's accumulated points, initialized to zero.
- **Lives:** The number of remaining attempts, initialized to three.
- **Time Remaining:** The amount of time left on the sixty-second countdown, initialized to sixty.

- **Rescue Counter:** The number of Baby Turtles successfully collected, initialized to zero.

Colliding with a vehicle causes the player to lose one. After losing a life, the Player Turtle returns to its starting position. Previously rescued Baby Turtles remain rescued.

Players should monitor their lives and remaining time throughout the game.

Win/Lose
To win Shell Out!, the player must successfully collect all three Baby Turtles and reach the Home location before the sixty-second countdown expires.

After reaching Home with all three children, the game displays a victory screen. The victory screen presents the player's final score and game statistics, recognizing the successful completion of the rescue mission. Score, along with Lives and Time Remaining, are displayed.

To lose, the player can meet two failure conditions:

1. **Running Out of Lives:** If the Player Turtle collides with vehicles enough times to lose all available lives, the game ends. A game-over screen appears, displaying the player's final score and game statistics.
2. **Running Out of Time:** The player has sixty seconds to rescue all three Baby Turtles and reach Home. If the countdown reaches zero before the rescue is completed, the Ravenous Pelicans enter the game world and steal any remaining Baby Turtles. This results in a game over. Similar to the former method, a game-over screen appears and displays the same statistics.

Both losing conditions prevent the player from playing further, and they must restart to play the game again and attempt the rescue mission once more.

References/Links

https://github.com/yataraou/-CSC170-Greenfoot-Project-ShellOut

[CSC170] Greenfoot Project Prospectus